Notes

Designing Large Scale AI Systems

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1 Introduction

1.1 Large Language Models (LLMs)

1.2 Software 3.0

The term "Software 3.0," introduced by Andrej Karpathy, describes how AI is reshaping the way we build software.

- **Software 1.0**: explicit rules written by programmers (if/else logic, loops).
- **Software 2.0**: behavior learned from data (neural networks trained on examples).
- **Software 3.0**: behavior specified in natural language (LLMs following prompts).

1.3 Roles of AI

Over the past decade, AI has moved from a niche technology to a core part of software design. In practice, AI systems tend to play three main roles:

- **AI as a Coding Agent**: AI systems can write, refactor, and review code, accelerating software development. Their output is largely deterministic because it is traditional code that can be inspected, tested, and debugged.
- **AI as a Component**: In this case, AI is embedded within a larger system, and the surrounding architecture is designed around its capabilities and limitations. This is not entirely new: many traditional machine learning models already fit this pattern.
- **AI as a User**: AI systems can also act as users of other systems, because they can consume inputs and generate outputs through your interfaces. Today, system design must account for AI clients that can call APIs, read documentation, and operate on application data. This is introducing a new question: Who are you building for? Human users, AI users, or both?

To design responsibly, we should distinguish when AI is unnecessary, when it is simply convenient, and when it is a true system component. Those choices are important since they respond to the question "What are you building for?" and this mainly depends on how reniewable and fixed we want our outputs.

- **No AI**: Use traditional software when rules are clear, logic is straightforward, and outputs are deterministic and easy to test.

- **AI as a Convenience**: The problem and rules are well understood, but prompting is faster or simpler than writing code. Outputs are still mostly predictable and testable, but reliability checks remain important.
- **AI as a System Component**: Use AI when behavior is too complex or context-dependent to encode with explicit rules. In these cases, conventional testing is harder because expected outputs are open-ended, and the value comes from handling ambiguity (for example, creative generation and open-ended Q&A).

This course focuses primarily on AI as a system component, while still covering the other two cases. In this context, we want to analyze how reviewable and fixed outputs are, and for which kind of users we are building for.

1.4 Challenges

Creating AI systems is more complex than traditional software because of the following challenges:

- **Getting a single call right**: LLMs are not deterministic, and their outputs can vary widely based on the prompt, which needs to be accurately designed to get the desired output.
- **Managing a conversation**: When building a system that interacts with an LLM, you need to manage the context across the conversation, which can make the costs grow rapidly.
- **Long-term memory**: LLMs have limited context windows, so you need to design strategies for storing and retrieving information (like preferences or conversation history) over time.
- **Enterprise/Community knowledge**: LLMs could use private data to train, so we need to design systems that can leverage this knowledge while respecting privacy and security concerns, also of our enterprise.

2 Prompting

Prompting is the process of designing inputs to LLMs in order to obtain the desired outputs. Effective prompting is essential when building AI systems because it directly affects output quality, reliability, and cost.

In this course, we focus on prompting through **API calls** rather than consumer user interfaces. The two settings differ significantly:

- **User Interfaces**: These are designed for end-users and often include features that help guide the user in crafting effective prompts, such as templates, examples, and interactive feedback. They usually don't allow you to access the model directly, so you have less control over the prompting process, but the interface itself can help users get better results.
- **API Calls**: When using APIs, you have more control and flexibility over the prompting process, as you are talking directly to the model. You have to design the prompt yourself, and you can also manage the conversation state and context more effectively. Modern APIs may also expose built-in tools such as web search, code execution, and retrieval, which can further improve results.

In short, API prompting is developer-centric, while user interfaces are user-centric. For instance, obtaining structured output can be difficult in many interfaces but straightforward through an API call.

2.1 Anatomy of a Prompt

A prompt typically consists of the following components:

- **System Prompt**: Sets the persona, rules and constraints the model follows throughout the conversation (job description). Conventionally, the system prompt is the same across all calls, but needs to be repeated since the API is stateless.
- **User Prompt**: The actual task or question you want the model to address. It can include the input data, examples and instructions on how to format the output. Conventionally, the user prompt is the part that changes across calls, depending on the specific task or input data.
- **Parameters**: These are settings that influence the model's behavior, such as temperature, top-p, max tokens, and stop sequences.

2.2 API Parameters

When making API calls to an LLM, you can specify parameters that influence model behavior:

- **Temperature**: Controls output randomness. Higher values (e.g., 0.8) usually produce more diverse and creative text, while lower values (e.g., 0.2) usually produce more focused and consistent text. In practice, the model samples from token probabilities, and temperature shapes that sampling.
- **Top-p**: Controls diversity by sampling only from the smallest set of tokens whose cumulative probability reaches `p`. It is typically used as an alternative to temperature; in many cases, tuning one is enough.
- **Max Tokens**: Caps response length, helping control both cost and verbosity.
- **Stop Sequences**: Define one or more strings that cause generation to stop when encountered.
- **Frequency Penalty**: Penalizes new tokens based on their existing frequency in the text so far, reducing repetition.
- **Presence Penalty**: Penalizes new tokens based on whether they appear in the text so far, encouraging the model to talk about new topics.

Beyond these parameters, other choices also influence output quality, especially **model selection** (for example, different model families) and **API type**.

It is useful to distinguish between **completions** and **responses**: completions are generally **stateless** (each call starts fresh unless you resend context), while responses can be **stateful**, with server-side conversation state that can simplify multi-turn interactions.

2.3 Prompting Strategies

This section covers common prompting strategies that improve model performance. Calibrating specifically is a key part of the design process, as overspecifying or underspecifying the prompt can lead to suboptimal outputs. Cutting redudancy and keeping a good structure permits to get better results, since the model follows literally the instructions

The best strategy to get a good prompt is to iteratively test and refine it, starting with a simple version and gradually adding complexity as needed.

2.3.1 Prompt Structure

Defining a clear **scope** for the model is crucial to prevent it from going off-topic or generating irrelevant information. To better handle complex tasks, it's also important to use **structure formatting** in the prompt, which can help guide the model's attention and improve the quality of the output. An example is to separate instructions, input data and output format into distinct sections, using clear headings or bullet points.
The suggested structure for a prompt is the following:

- **Role (optional)**: Define the role or persona the model should adopt (for example, "You are a helpful assistant"). This differetiates the **direct instruction** prompting strategy, where you simply provide clear, literal instructions for simple tasks, from the **role prompting** strategy, where you shape the tone, vocabulary and depth of the response.
- **Goal**: Clearly state in one sentence the overall goal or task (for example, "Summarize the following text in one sentence"). (This last part doesn't improve factual correctness)
- **Context**: Provide any necessary background information, data, or examples that the model needs to complete the task.
- **Constraints**: Specify any rules like audience, length limits and tone.
- **Output Format**: Define the expected format of the output (for example, "Return a JSON object with the following fields...").
- **Verify**: Ask the model to check its own output for errors or inconsistencies, which can help improve reliability.

Also **where you place instructions** relative to data significantly affects the quality, especially with long data. The common rule is to place long data first, instructions last and query at the very end.

2.3.2 Few-shot prompting

Few-shot prompting involves providing the model with a few examples of the desired output format or behavior within the prompt itself. This can help the model understand the task better and generate more accurate responses.
It's important to choose diverse examples covering typical cases and edge cases, and keep a consistent format across examples to help the model learn the pattern. Also ordering examples matters: place the most similar examples last, closer to the actual task, as the model tends to pay more attention to recent context.

2.3.3 Chain-of-thought prompting

LLMs generate tokens sequentially. When you force intermediate reasoning steps, each subsequent token is conditioned on better context. In other words, the model "thinks" by writing, and by prompting it to write down its reasoning process, you can often improve the quality of the final answer, especially for complex tasks that require multiple steps of reasoning.
There are several variations of this strategy:

- **Zero-shot CoT**: You simply ask the model to explain its reasoning process before giving the final answer, without providing any examples.
- **Few-shot CoT**: You provide examples of the reasoning process along with the final answer, which can help the model learn how to structure its reasoning.
- **Structured CoT**: You explicitly define the intermediate steps or subtasks that the model should follow, which can help guide its reasoning process more effectively.

3 Stateful Agents

The models discussed thus far are **stateless agents**. This means they do not maintain a memory of previous interactions; instead, they treat every request as an independent event and respond based solely on the current input.

To illustrate this, imagine you have access to an infinite sequence of *superhumans*. You approach each one and ask exactly one question. While each superhuman provides a brilliant response to that specific query, they possess no knowledge of your previous conversations with their predecessors. Every time you move to a new person, the context is reset to a blank slate. This behavior characterizes a stateless system.

In contrast, a **stateful agent** retains information across multiple turns. It stores previous inputs and outputs—often referred to as the *conversation history* or *context*—allowing it to understand follow-up questions and maintain continuity in a dialogue.

In this section we will explore various techniques used in agents design, from simple stateless agents to more complex stateful ones.

3.1 Stateless

As mentioned, stateless agents do not maintain any memory of past interactions. The `chat completition` api is an example of a stateless agent by design.

```
# Turn 1: tell the model your name
r1 = client.chat.completions.create(
  model="gpt-4.1-mini",
  messages=[{"role": "user", "content": "My name is Alice."}],
)
# Turn 1: "Nice to meet you, Alice!"

# Turn 2: ask for your name  but we DON'T send turn 1
r2 = client.chat.completions.create(
  model="gpt-4.1-mini",
  messages=[{"role": "user", "content": "What is my name?"}],
)
# Turn 2: "I don't have access to your name."
```

The model can't answer because it literally never saw the first message.

Although this type of agents seems pretty useless, they can be fine for single-shot tasks, like writing a poem, generating an image, or answering a question.

3.2 Default Stateful Agents

Open AI `Response API` can be used to create stateful agents.

```
resp1 = client.responses.create(model="gpt-4.1-mini", input=
    "My name is Alice.")

resp2 = client.responses.create(
  model="gpt-4.1-mini",
  input="What is my name?",
  previous_response_id=resp1.id,   # server remembers turn
      1
)
# "Your name is Alice."
```

This is convenient, but it's a black box approach. We have no control over what the model remembers and how it uses that information.

All the next techniques we will manage the state ourselves, giving us more control and flexibility.

3.3 Full History

A first approach to create a stateful agent is to send with each request the full conversation history. This is obviously the most straightforward way to maintain context, but it can quickly become inefficient as the conversation grows longer. In fact, each time we send a new message, the number of tokens we need to send increases quadratically, since we need to resend all previous messages.

This is not only inefficient in terms of latency and cost, but it can also lead to fulling the model's context window, which can cause lower quality responses or even errors.

3.4 Sliding Window

To mitigate the issues of the full history approach, we can use a **sliding window** technique. Instead of sending the entire conversation history, we only send the most recent N messages, where N is a predefined window size. This way, we can maintain some context while keeping the number of tokens manageable and constant over time.

The downside of this approach is that we might lose important information from earlier in the conversation, which can lead to less coherent responses if the model relies on that information.

3.5 Summarization

To address the limitations of the sliding window technique, we have to find a way to retain important information from the conversation without sending the entire history. This is where **summarization** comes into play. The idea is to periodically summarize the conversation history and send the summary along with the most recent messages. This way, we can keep the context manageable while still retaining important information from earlier in the conversation.

The summarization is usually done by the model itself. Along with the user message, we can provide the model with the current summary and ask it to update the summary based on the new message.

This way, the memory stays bounded, without the loss of information.

A few tips for effective summarization prompts:

- Use a structured format for the summary, organizing information into categories (e.g., "Facts", "Opinions", "Actions", etc.);
- Set a maximum length for the summary to ensure it remains concise and focused on the most relevant information;
- Resolve contradictions in the summary by asking the model to identify and address any conflicting information.

3.6 Long-Term Memory

Every technique we have seen so far is based on the idea of a *working memory*, that dies whenever the conversation ends. However, in many applications we want to have a *long-term memory* that persists across multiple conversations and sessions.

This is usually achieved by:

- Once the conversation ends, extract important information. Ask the LLM to pull out structured facts worth remembering: facts, preferences, decisions, expertise, etc;
- Store this information in an external database;
- When a new conversation starts, retrieve relevant information from the database and inject it into the prompt.

This is the actual pattern used behind the OpenAI `ChatGPT` memory feature.

3.7 When to use which pattern

Each strategy can be used in different scenarios, depending on the specific requirements of the application and the constraints of the model being used.

- **Stateless**: suitable for single shot tasks (e.g., translate, classify);
- **Full History**: suitable for short conversations;
- **Sliding Window**: suitable for longer conversations where only the most recent context is relevant;
- **Summarization**: suitable for long conversations where important information needs to be retained without exceeding the context window;
- **Long-Term Memory**: suitable for applications that require personalization or have returning users, where information needs to persist across sessions.

4 Agent Tools

LLMs by themselves are very powerful when it comes to understanding and generating human language. However, they can be even more powerful when combined with external tools that allow them to perform specific tasks or access up-to-date information. In this section, we will explore some of the tools that can be integrated with LLMs to enhance their capabilities.

They enable a wide range of applications, such as:

- **Computational tools:** perform complex calculations, data analysis, and other computational tasks that may be beyond their native capabilities;
- **Knowledge retrieval tools:** access external databases, APIs, or the internet to retrieve up-to-date information;
- **Action execution tools:** interact with other software or hardware to perform actions based on the LLMs output.

4.1 Register a Tool

LLM providers usually offers ready-to-use tools, such as a web search tool, code interpreter, file search, and more. However, they also allow users to create and register custom tools that can be tailored to specific use cases or domains.

To register a tool, we first need to define its schema, which includes the tool's name, description, and input parameters. An example of a tool schema is shown below:

```
{
"type": "function",
"function": {
    "name": "explore_data",
    "description": "Inspect dataset schema, missing values, and candidate targets. Call first when the dataset structure is unknown.",
    "parameters": {
    "properties": {
        "sample_rows": {"type": "integer", "description": "Number of sample rows to return"}
    },
    "required": ["sample_rows"]
    }
}
}
```

4.2 Multi-Tool loop

Real world problems often require the use of multiple tools in a sequence, where the output of one tool serves as the input for another. The model must reason about which tool to call next based on what it's learned so far.

Example: Data Analysis Agent Consider the following example, of a data analysis agent. The following tools are registered:

- **Explore Data:** inspects the dataset schema, missing values, and candidate targets. it has no preconditions;
- **Cluster Data:** performs clustering on the dataset, discovering natural groupings in the data. Needs a numeric column as input;
- **Classify Data:** performs regression or classification on the dataset, depending on the target variable. Needs a target column as input.

Different queries may require different sequences of tools. For example:

- **Query 1:** "What kind of data do I have?" → Explore Data;
- **Query 2:** "Are there any natural customer segments in the data?" → Explore Data → Cluster Data;
- **Query 3:** "Segments and churn prediction for age=29, income=50k..." → Explore Data → Cluster Data → Classify Data.

4.3 Problems with tool use

The usual problem with multi-tool agents is that they can get stuck in loops, calling the same tool repeatedly without making progress towards the final goal. To mitigate this issue, we can implement a control mechanism that allows the use of a tool if and only if the following conditions are met:

1. Some part of the original query is still unanswered;
2. Another available tool can resolve it;
3. The tool call is necessary, not just helpful.

This should help the agent avoid common pitfalls, such as calling tools just for the sake of calling them, or getting stuck in a loop with no progress.

Another approach, particularly useful when queries are complex or ambiguous, is to allow the agent to ask for clarification from the user before deciding which tool to call next.

Keep in mind that how we design the tools and their interfaces can also influence how effectively the agent uses them. In fact, the model picks them based on their descriptions and parameters. Therefore, it's important to make them as clear and informative as possible, to help the model understand when and how to use them effectively.

5 Evaluation

In this section, we will discuss the basic recipe for "evaluating" in different periods of software development, starting from the early days of software development, to the current era of Software 2.0 and 3.0, focusing on the differences in the evaluation process and the challenges that arise in each era.

5.1 Software 1.0 - Evaluation in Traditional Software Development

In traditional software development, the evaluation process is relatively straightforward. The expected output of the software is well-defined, and we can easily compare the actual output with the expected output to determine if the software is working correctly. This process is often referred to as "testing", and it can be done through:

- **Unit Testing:** testing individual components or functions of the software to ensure they work as expected. For example, if we have a function that adds two numbers, we can test it by providing different pairs of numbers and checking if the output is correct;
- **Integration Testing:** testing the interaction between different components of the software to ensure they work together correctly;
- **Regression Testing:** testing the software to ensure new changes do not break existing functionality.

The process is **deterministic** and **binary**, meaning that the software either works or it doesn't, and the output is always the same for the same input. The only difficulty in this process comes from the coverage of the tests, not from the evaluation itself.

5.2 Software 2.0 - Evaluation in Machine Learning

In the era of Software 2.0, where machine learning models are used to perform tasks, the evaluation process becomes more complex. This is because the output of the model is **not deterministic** anymore, but rather **probabilistic**, The evaluation process, also called **experimentation**, is no longer binary, but rather a matter of degree. It shifts from "is the output correct?" to "how good is the model across a test set?".

For our analysis, we will focus on the evaluation of a binary classification model, where the goal is to classify inputs into two categories (for example, "spam" vs "not spam", or "positive" vs "negative"). Usually the metrics used is the proportion of misclassified inputs (also known as **error**), meanwhile the proportion of correctly classified inputs is called **accuracy**.

5.2.1 Comparing Models

Before putting a model into production, we want to compare it with other models to understand which one performs better on a given task. We could also compare different versions of the same model, to understand if the changes we made have improved the performance or not.

In simple cases, such as in 2D dataset, we can plot the predicted output and visually inspect the decision boundary to answer these questions. However, in more complex cases, such as in high-dimensional datasets, we need to rely on quantitative metrics to compare the models.

The most straightforward approach is to train both models on the same training set, and pick the one with fewer mistakes. This generates a few questions:

- On which data we compute the error?
- How we eliminate random effects?
- Are the metrics we are using good?

5.2.2 Data Splitting

As you have surely heard before, you should **never judge your performance on the training set**, since the model has already seen them and can easily memorize them, leading to an overestimation of the model's performance on unseen data.

Test Set

The first thing that should be always done in machine learning is to split the original dataset into a **training set** and a **test set**. The training set is used to train the model, while the test set is used to evaluate the model's performance on unseen data, which is more representative of the real-world performance of the model.

How should we split the dataset? The only factor to consider when splitting the dataset is the **size** of the **test set**. The bigger it is, the more it is representative of the real-world data distribution, and the more reliable the evaluation results will be.

Ideally, we want 10 000 examples in the test set, but this is not always possible, especially if the original dataset is small.

Overfitting the test set

However, this approach can lead to overfitting on some cases. Suppose we want to choose the best hyperparameters for our model, and we are provided with a test set. We try different hyperparameter settings, and we pick the one that performs best on the test set, supposing that it will also generalize well to unseen data.

At a certain point, we are provided with a fresh test set, and we find out that the performance is much worse than expected with the chosen hyperparameters. This is because we have **overfitted the test set**, meaning that we have chosen hyperparameters that perform well on the test set, but not on unseen data. Our method for choosing the best hyperparameters are another learning algorithm, and by testing different hyperparameters on a small test set, we are essentially training this learning algorithm on the test set, leading to overfitting.

We can see this as a form of `multiple testing` problem in statistics. We are testing multiple hypotheses (different hyperparameter settings) on the same test set, and we are picking the one that performs best, which is likely to be a false positive.

The simple answer to this problem is to **never use the same test set more than once**, and to always use a fresh test set for the final evaluation of the model's performance.

Validation Set

Reusing the test set not only makes you choose the wrong model, but also **inflates the performance measures**, making them look better than they actually are.

To avoid this, we need to choose the best best model and/or hyperparameters on a separate set of data, extracted from the training set and called the **validation set**. Ideally, the validation set is the same size of your test set, but you can make it a little smaller to get more data for training.

Note that this approach alone does not solve the problem of multiple testing, but it just provides a final fail-safe to detect it, You need always to be careful and whenever it's needed to rerun your experiments on training and validation sets, as you can easily end up overfitting the validation set as well.

However, if your dataset is large, and you haven't done anything strange in tuning phase, the validation set results should be a good estimate of the test set results, and you can be reasonably confident that the model will perform well on unseen data.

Cross Validation

In some cases, especially when the dataset is small, we can't afford to set aside a large portion of the data for testing and validation, since we need as much data as possible for training.

A technique that can help in this case is called **Cross validation**. It allows us to use the entire dataset for both training and testing, by splitting it into multiple folds. For example, in a 5-fold cross validation, we split the dataset into 5 equal parts, and we train the model on 4 parts and test it on the remaining part (validation). We repeat this process 5 times, each time using a different part as the test set,and we average the results to get a more reliable estimate of the model's performance.

This can be time-consuming, but ensures that every instance in the dataset is used for training. After chosen your model and hyperparameters through the cross-validation process, you can test it once on the test set to get an estimate of its performance on unseen data.

A special case: Temporal Data

If your data has a temporal ordering, you need to take it into account when splitting the dataset. In such cases, you can't sample your test set randomly, since training on future data and testing on past data is not a realistic scenario. The ordering in the dataset needs to be maintained when splitting the dataset, to ensure that the model is trained on past data and tested on future data, which is more representative of the real-world scenario.

However, this is not always the case. If your data has a timestamp, but there's no meaning associated with the ordering of the data (such as in email classification), you can still split the dataset randomly, since the ordering does not provide any useful information for the task at hand.

If you want to apply cross validation to temporal data, that takes a small portion of initial data as the training set and then iteratively adds more data to the training set, while validating on the next portion of data, until you reach the end of the dataset.

Final remark: Philosophically talking, when in doubt make sure that the evaluation setting accurately simulates production, and that the validation setting accurately simulates the evaluation setting,

5.2.3 Experiment Design

Now that we now how to experiment, what experiments should we run? Which hyperparameters should we test? Basically as long as you are not looking at the

test set, you can do whatever to like.

Hyperparameter Tuning

When tuning hyperparameters, we have the following options:

- **Trial and Error**: This is the most basic approach, where you simply try different combinations of hyperparameters and see which one performs best on the validation set. This is driven by intuition and experience, and it can be time-consuming, especially if you have a large number of hyperparameters to tune.
- **Grid Search**: This is an automated approach, where you define a grid of relevant hyperparameter values, and the algorithm trains and evaluates a model for each combination of hyperparameters in the grid. This can be computationally expensive, especially if the grid is large, but it ensures that you are exploring a wide range of hyperparameter combinations.
- **Random Search**: This is another automated approach, where you randomly sample combinations of hyperparameters from a defined distribution, and train and evaluate a model for each combination. This can be more efficient than grid search, especially if you have a large number of hyperparameters, since it allows you to explore a wider range of combinations.

5.3 Software 2.0 - Statistical Testing

Machine Learning and statistics are closely related, but significance testing is rarely used in ML experiments due to large sample sizes and emphasis on replication over statistical significance. Replication is key, but often challenging in practice, so we can still use statistical testing as a tool to assess the reliability of our results.

5.3.1 True metrics vs Estimated metrics

When we evaluate a model, we usually imagine data being sampled from a probability distribution $p(x)$. What we really want is to develop a classifier that performs well on any data point sampled from the original distribution.

Imagine for example we sample one data point from the distribution, and we classify it with some classifier c. The probability of c making a correct prediction on this data point is the **true accuracy**. However, we can't compute the true accuracy, since we don't have access to the entire distribution, but only to a finite sample of data points.

Instead, what we usually do is to compute the so-called **sample accuracy**. We take a large enough sample of data points from the distribution, and we compute the accuracy of the classifier on this sample, which is an estimate of the true accuracy.

Now, given the estimated sample accuracy, the question is: how sure can we be that the true accuracy is close to the sample accuracy?

Confidence Intervals

One way to quantify the uncertainty around our estimate is to compute confidence intervals, which are ranges of values that are likely to contain the true metric with a certain level of confidence. This helps us understand the uncertainty around our estimate and make more informed decisions about model performance.

The confidence interval is computed based on the variability of the sample metric across different samples, and it gives us a range of values within which we can be reasonably confident that the true metric lies.

The size of the confidence interval depends on two factors:

- **True metric**: If the true metric is close to 0 or 1, the confidence interval will be narrower, since there is less variability in the sample metric.
- **Sample size**: The larger the sample size, the narrower the confidence interval, since there is less variability in the sample metric across different samples.

If you have not the luxury of a large sample size, you can do some statistical testing to understand how much the sample metric is close to the true metric.

Remark: Since we are working with estimated values, the confidence interval itself is an estimate. It is not guaranteed to contain the true metric with the specified confidence level for any single realization. The interval is relative to the specific test sample collected; therefore, it is not a fixed range but one that varies across different samples.

Standard Deviation and Standard Error

The **standard deviation** (σ) measures the amount of variation or dispersion within a set of data values. In the context of a sampling distribution, the region spanning approximately two standard errors to the left and two to the right of the sample mean (a total range of four standard errors) contains roughly 95% of the sample metrics across different samples. Consequently, this interval

$[\bar{x}-2SE, \bar{x}+2SE]$ represents a 95% **confidence interval** for the true population metric.

The **standard error** (SE) is the standard deviation of the sampling distribution of a statistic, most commonly the sample mean. It quantifies the uncertainty of the sample estimate. It is computed as:

$$SEM = \frac{\sigma}{\sqrt{n}}$$

Where σ is the standard deviation of the individual observations and n is the sample size.

While the standard deviation is a measure of the **spread** of the underlying data, the standard error and the confidence interval are measures of **uncertainty** regarding the true population parameter. Increasing the sample size n will typically decrease the standard error, thereby narrowing the confidence interval, even if the underlying standard deviation of the population remains constant.

5.4 Randomness

In machine learning, the results of an experiment can vary significantly upon rerunning due to **random initialization** of model weights and stochastic optimization processes. Additionally, **data sampling** introduces randomness because the available dataset is merely a single realization of a larger, underlying population.

To evaluate robustness to random initialization, executing the experiment multiple times with different seeds is often sufficient. To test robustness to data sampling, we typically employ the approaches described in the following sections.

Resampling Methods

To assess how much our evaluation metric might fluctuate due to data sampling, we rely on **resampling methods**. The core idea is to repeatedly recompute the sample metric on different subsets of the data, allowing us to estimate its standard deviation and construct confidence intervals for the true metric.

For instance, in cross-validation, we compute the metric on each splitted fold. By analyzing how much the metric varies across these folds, we gain insight into the robustness of our results and the degree to which they depend on the particular sample at hand.

However, as we increase the number of repetitions, especially with small datasets, the overlap between samples also increases, which can limit the independence of our estimates.

Bootstraping Methods

An alternative approach is to use **bootstrapping methods**, where we repeatedly **resample with replacement** from the original dataset to generate multiple new samples, called bootstrap samples. Each bootstrap sample is the same size as the original dataset, but because sampling is done with replacement, some data points may appear multiple times or not at all.

This technique allows us to create as many resampled datasets as needed, without being limited by overlap concerns. By evaluating our metric on each bootstrap sample, we can estimate its variability and construct confidence intervals, often with greater flexibility than traditional resampling methods.

In practice, bootstrapping provides a powerful way to approximate the underlying data distribution.

5.5 Evaluating Regression

When evaluating regression models, we need to compute continuous metrics such as the **mean squared error** (MSE). Here, we assume that each observation is drawn from a distribution centered around some true underlying value. The sample mean, calculated from our test results, provides a point estimate of this true metric. However, it is important to remember that the sample mean itself is a random variable, and it will vary from sample to sample.

For small sample sizes, the distribution of the sample mean follows a Student's t-distribution, which approaches a normal distribution as the sample size increases ($n > 30$), both centered at the true mean. This statistical framework enables us to construct confidence intervals for the true metric, using the sample mean and its observed variability across samples.

Other metrics for regression include the **mean absolute error** (MAE), which measures the average absolute difference between the predicted values and the true values, and the **root mean squared error** (RMSE), which is the square root of the MSE and provides an interpretable metric in the same units as the original data.

5.5.1 Bias and Variance

Bias and variance are two fundamental sources of error in regression models, and their definitions are closely tied to the mean squared error (MSE), though they can be extended to other metrics as well.

- **Bias**: This is the difference between the optimal MSE (the lowest possible error for your model class) and the true MSE. Bias reflects how much your model is systematically underfitting the data. Importantly, bias remains

unchanged even if you resample the data, as it is an inherent property of the model itself, not the data.

- **Variance**: This is the difference between the true MSE and the observed MSE across different samples. Variance captures how much your model's predictions fluctuate due to variations in the training data. Unlike bias, variance can change from sample to sample, since it depends on the data rather than the model.

Typically, there is a trade-off between bias and variance: increasing model complexity tends to reduce bias but increase variance, while simpler models have higher bias and lower variance.

5.6 Evaluating Classification

When evaluating classification models, we rely on metrics such as accuracy, precision, recall, and F1-score. These metrics are derived from the confusion matrix, which summarizes the counts of true positives, true negatives, false positives, and false negatives.

5.6.1 Confusion Matrix

The confusion matrix summarizes a classification model's performance by counting true positives, true negatives, false positives, and false negatives. From this table, we can compute a variety of metrics to evaluate the model, for example the **accuracy**, which is the proportion of correctly classified instances. While it can be challenging to directly compare two models using only their confusion matrices, this tool provides valuable insights into the types of errors a single model makes.

Application of Confusion Matrix

Plotting the confusion matrix for the training, validation, and test sets provides valuable insights into model behavior and helps diagnose issues like overfitting or poor generalization.

Typically, we use **test accuracy** to assess the validity of our conclusions, **validation accuracy** to tune hyperparameters, and **training accuracy** to compare against validation accuracy (the **generalization gap**) to detect overfitting or underfitting. Training accuracy alone should not be used for model evaluation.

5.6.2 Imbalance Issues

Imbalance issues arise when classes are not equally represented or have different importance.

Class imbalance occurs when one class is much more frequent than the other, making high accuracy misleading. For example, a model may simply predict the majority class and still achieve a high score. If the minority class is under-represented in the training set, the model may fail to learn about it, resulting in poor generalization to new data.

Furthermore, the cost of misclassifying one class (**cost imbalance**) can be much higher than the other (e.g., failing to detect a sick patient is typically worse than a false alarm). In such cases, it is important to use metrics and error calculations that reflect both class imbalance and cost imbalance, often by weighting errors according to their impact based on the specific application.

5.6.3 Precision and Recall

Precision measures the proportion of predicted positives that are actually correct, while **recall** measures the proportion of actual positives that are correctly identified.

$$Precision = \frac{TP}{TP + FP} \qquad Recall = \frac{TP}{TP + FN}$$

There is often a trade-off between precision and recall, especially in the presence of class imbalance: increasing one can decrease the other. Plotting precision and recall for different classifiers or thresholds helps visualize and manage this trade-off.

5.6.4 ROC Space: True Positive Rate vs False Positive Rate

TPR (True Positive Rate) measures the proportion of actual positives correctly identified, while **FPR** (False Positive Rate) measures the proportion of actual negatives incorrectly classified as positives.

$$TPR = \frac{TP}{TP + FN} \qquad FPR = \frac{FP}{FP + TN}$$

The trade-off between maximizing TPR and minimizing FPR is visualized in the **ROC curve** (Receiver Operating Characteristic), where the ideal point is TPR = 1 and FPR = 0. The ROC curve helps compare classifiers across different thresholds.

5.6.5 Plotting thresholds

So far, we have discussed comparing different models, but we can also evaluate a single model by varying its decision threshold. This analysis is meaningful and possible only if the model is a **ranking classifier**, that is, it outputs a continuous score or probability for each instance. For ranking classifiers, **ranking error** refers to cases where the model assigns high confidence to incorrect predictions (e.g., predicting positive with high confidence when the true label is negative).

For instance, when using a high threshold ("shy" model) results in few positive predictions, while a low threshold ("bold" model) predicts more positives, increasing the risk of false positives. Plotting performance metrics (such as ROC or Precision-Recall curves) across thresholds helps visualize this trade-off and select the optimal threshold for a specific application.

5.6.6 Coverage Matrix

For ranking classifiers, we can visualize their predictions using a **coverage matrix**. In this plot, the x-axis represents the confidence for the negative class, and the y-axis represents the confidence for the positive class.

Points near the top right corner show strong confidence in predicting positive, while points near the bottom left show strong confidence in predicting negative. By coloring each point according to the predicted class (after applying a threshold), we get an intuitive visual summary of how well the model separates the two classes.

5.6.7 Coverage Matrix

For ranking classifiers, a **coverage matrix** visualizes all possible pairs of positive (rows) and negative (columns) examples. Each cell compares a positive example x^+ and a negative example x^- using the model score $s(\cdot)$:

- **Green cell**: $s(x^+) > s(x^-)$ — the model correctly ranks the positive above the negative.
- **Red cell**: $s(x^+) \leq s(x^-)$ — the model ranks the pair incorrectly (ranking error).

The proportion of red cells is the probability of making a ranking error.

By introducing a threshold τ on $s(\cdot)$, we turn the scores into binary predictions: $s(x) \geq \tau$ is classified as positive. Varying τ changes which pairs are correctly ordered, linking the coverage matrix to the ROC curve.

5.6.8 Area Under the Curve (AUC)

The **AUC** (Area Under the Curve) is a single metric that summarizes a model's performance across all possible thresholds. It is calculated as the area under the ROC curve and is widely used to compare different classifiers.

Intuitively, the AUC represents the probability that a ranking classifier assigns a higher score to a randomly chosen positive example than to a randomly chosen negative one. In other words, it quantifies how well the model ranks positive examples above negative ones, regardless of the specific threshold used for classification. In terms of the coverage matrix, the AUC is the proportion of green cells (correctly ranked pairs).

Partitioning Classifiers For some models, such as decision trees (**partitioning classifiers**), scores are assigned based on regions or leaves. If we use the proportion of positive points in each leaf as the score, all points in the same leaf share the same score. This can result in ties, where some pairs cannot be strictly ranked. For large datasets, these ties have less impact on the overall AUC.

5.6.9 Precision-Recall Curve

The **Precision-Recall (PR) curve** is another tool for evaluating classification models. Instead of plotting TPR vs. FPR (as in the ROC curve), the PR curve plots precision against recall for different thresholds.

The area under the PR curve (AUC-PR) provides a single-number summary, which is especially informative when dealing with imbalanced datasets—situations where the ROC curve may be overly optimistic.

However, the PR curve is not normalized and can be harder to interpret than the ROC curve, as its baseline depends on the proportion of positive examples. Both curves offer complementary insights, so it is often helpful to plot and compare them together.

5.7 No Free Lunch Theorem

It is natural to ask: which classifier, model, or search method is the best? The **No Free Lunch Theorem** tells us that, without knowing the specific problem, there is no universally best algorithm.

If we average over all possible problems, every optimization or learning algorithm performs equally well. An algorithm only excels when its assumptions match the problem at hand. This is known as the **problem of induction**. Every learning algorithm or model has an **inductive bias**: a set of built-in assumptions about the data. Inductive bias determines how a model generalizes from training

data to unseen data. For example, a linear model assumes linear relationships between features and the target variable.

As practitioners, our task is to choose the inductive bias that best fits our problem. When two models explain the data equally well, we usually prefer the simpler one, following the principle that simple patterns are more likely than complex ones. This is why many learning methods include a built-in **bias toward simplicity**.

6 Statistical Testing

ML and statistics are closely related, but significance testing is rarely used in ML experiments due to large sample sizes and emphasis on replication over statistical significance. Replication is key, but often challenging in practice, so we can still use statistical testing as a tool to assess the reliability of our results.

6.1 True metrics vs Estimated metrics

When we evaluate a model, we usually imagine that data a sampled from $p(x)$, and we want a classifier that minimizes the error on the true distribution, not the one we have in our dataset. However, we can only compute the statistic on the dataset, which is called the **sample metric**. All metrics have a true value defined on the true distribution, but we can only compute an estimate of it based on the data we have.

This brings to us the need to prove how much near the sample metric is to the true metric, and that is not just a noise.

6.1.1 Confidence Intervals

One way to do this is to compute confidence intervals, which are ranges of values that are likely to contain the true metric with a certain level of confidence. This helps us understand the uncertainty around our estimate and make more informed decisions about model performance.

This can be done using various test samples, making the sample metric a random variable. This makes the sample metric vary across different samples, and we can use this variability to compute confidence intervals.

The size of the confidence interval depends on two factors:

- **True metric**
- **Sample size**

If you have not the luxury of a large sample size, you can do some statistical testing to understand how much the sample metric is close to the true metric.

Remark: Since we are working with estimated value, also the confidence interval is an estimate, and it is not guaranteed to contain the true metric with the specified confidence level. So it relative to the test sample we have, and its important to understand that the confidence interval is not a fixed range, but it can vary across different samples.

Standard Deviation and Standard Error: The standard deviation of the sample metric is a measure of how much the sample metric varies across different samples. The standard error is the standard deviation of the sample mean, which is the sample metric in this case, and it is computed as the standard deviation divided by the square root of the sample size.

The standard deviation is the measure of **spread** of the sample metric, meanwhile the **confidence interval** and standard error are measures of **uncertainty** around the true metric. This makes the two concepts different.

6.2 Randomness

In ML we know that, the results of an experiment can completely change by rerunning due to the **random initialization** of the model. Also **data sampling** can introduce randomness, due to the fact that we are working with a sample of data, and not the entire population.

This is why we need to evaluate how much the results are robust to randomness. For random initialization, it's enough to run the experiment multiple times, but how we test the robustness to data sampling?

6.2.1 Resampling Methods

The basic approach is to analyze the **standard deviation** of the sample metric across different samples, and to compute confidence intervals for the true metric. Through cross-validation, we can compute the sample metric on different subsets of the data, and analyze how much it varies across these subsets. This allows us to assess the robustness of our results to data sampling, and to understand how much the sample metric is likely to vary if we were to collect a different sample of data.

However, as we increase the number of repeats, we are incrementing the overlap between the samples, especially if we are working with a small dataset.

6.2.2 Bootstraping Methods

An alternative approach is to use bootstraping methods, in which we **resample with replacement** from the original dataset to create multiple bootstrap samples. This allows us to sample a dataset that is exactly the same size as the original dataset, permitting us to resample as many times as we want without worrying about overlap between samples.
This method approximates the data distribution in a very precisely way.

6.3 Evaluating Regression

When computing a continuous metric, such as the mean squared error m, we assume individual observations are distributed around a true metric. The sample mean (computed from the results of the testing) serves as a point estimate of this expectation, but it is itself a random variable. Across different samples, the sample mean follows a Student's t-distribution which converges to a normal distribution for sample sizes $n > 30$, centered at the true mean. This allows us to compute confidence intervals for the true metric based on the sample mean and its variability.

6.3.1 Metrics

The most common metrics for regression is the mean squared error (MSE), which measures the average squared difference between the predicted values and the true values. However, also RMSE (root mean squared error) is used, which is the square root of MSE, and it is more interpretable since it is in the same units as the original data, instead of being in squared units.

6.3.2 Bias and Variance

Bias and variance are two sources of error in regression models. Those definitions are strictly related to the MSE, but they can be extended to other metrics as well.

- **Bias** It's the difference between your optimal MSE and the true MSE, and it is a measure of how much your model is underfitting the data. This part remains the same also if you resample the data, since it is a property of the model itself, and not of the data.
- **Variance** It's the difference between the true MSE and the measure (across samples), and it is a measure of how much your model is overfitting the data. This part can vary across different samples, since it is a property of the data, and not of the model itself.

Often the bias and variance are in a trade-off, since increasing the complexity of the model can reduce the bias but increase the variance, and vice versa.

6.4 Evaluating Classification

When evaluating classification models, we often use metrics such as accuracy, precision, recall, and F1-score. These metrics are based on the confusion matrix, which counts the number of true positives, true negatives, false positives, and false negatives.

6.4.1 Class Imbalance

It can happen that the classes in the dataset are imbalanced, meaning that one class is much more frequent than the other. In this case, getting a accuracy of 90% can be **misleading**, since the model can simply predict the majority class and still achieve a high accuracy.
In production, this can lead to a model that performs poorly on the minority class, which can be critical in certain applications, such as fraud detection or medical diagnosis.

Another problematic case, could be the one where all the minority class is contained in the test set, and the model is not able to learn anything about it, since it is not present in the training set. This can lead to a model that performs very poorly on the test set, and it is not able to generalize to new data.

6.4.2 Cost Imbalance

In some cases, the cost of misclassifying one class can be much higher than the cost of misclassifying the other class. For example, in medical diagnosis, misclassifying a sick patient as healthy can have much more severe consequences than misclassifying a healthy patient as sick.
In these cases, it's important to use metrics that take into account the class imbalance and cost imbalance. We can tune our error by multiplying the error of each class by a cost factor, which can be determined based on the specific application and the relative importance of the classes.

It's important to note that, in both cases, the cost is never zero, even in the example of medical diagnosis, since misclassifying a healthy patient as sick can lead to unnecessary treatments and anxiety for the patient. Especially in social applications, where could bring to a lot of ethical issues.

6.4.3 Confusion Matrix

The confusion matrix is a table that summarizes the performance of a classification model by counting the number of true positives, true negatives, false positives, and false negatives. This allows us to compute various metrics.

However it's difficult to compare two models based on the confusion matrix, but it can give good insights on the single model works. The problems previously described affects the diagonal of the confusion matrix (false positives and false negatives - misclassifications)

You can plot the confusion matrix both for the training set, validation set, and test set, giving different insights on the three sets.

6.4.4 Precision and Recall

Precision is the ratio of true positives to the total number of predicted positives, while **recall** is the ratio of true positives to the total number of actual positives. In order to handle class imbalance, we need to trade-off between precision and recall, since increasing precision can lead to a decrease in recall, and vice versa.
To help us in this trade-off, we can plot the precision and recall for different classifier.

6.4.5 TPR and FPR - ROC space

TPR (True Positive Rate) is the ratio of true positives to the total number of actual positives, while **FPR** (False Positive Rate) is the ratio of false positives to the total number of actual negatives. We want to maximize TPR and minimize FPR, but as before there is a trade-off between the two. The optimal space is TPR/FPR = 1, which means that we are maximizing TPR and minimizing FPR at the same time, this is called the **ROC space** (Receiver Operating Characteristic).

6.4.6 Plotting thesholds

Until now we have been talking about the comparison of different models, but we can also compare the same model with different thresholds (eager to predict positive). If we have a shy model (high threshold), we would not predict anything, however if we have a bold model (low threshold), we would predict everything as positive even risking misclassifications. The curve that we have, it can be a good way to compare the performance of the model across different thresholds, and to choose the optimal threshold based on the specific application and the trade-off between precision and recall.

However, this is possible only if the model outputs is **ranking classifier** (i.e., it provides a score for each instance), which is not always the case, since some models output a binary prediction without a score. In this case, the **ranking error** is the error measured in confidence, if we predict a positive class with a high confidence, but it is actually a negative class, this would be an high ranking error.

6.4.7 Coverage Matrix

Given a model that outputs a ranking/score, we can plot the coverage matrix, with negative class on the x-axis and positive class on the y-axis. Basically the more confident the model is predicting a positive class, the closer the point is to the top right corner, while the more confident the model is predicting a negative

class, the closer the point is to the bottom left corner. Then coloring binary the points based on the predicted class (after thesholding), we can have a good visual representation of the performance of the model.

6.4.8 AUC

The AUC (Area Under the Curve) is a metric that summarizes the performance of a classification model across all possible thresholds. It is computed as the area under the ROC curve, and can be used to compare different models. If we normalize the coverage matrix, we get the ROC curve, and the AUC is the area under this curve. The AUC (in ROC space) is an estimate of the probability that a ranking classifier puts a randomly drawn pair of positive and negative examples in the correct order.

Lets now look how to works for another type of classifier. The decision tree is an example of a **partitioning classifier**, and how we get a ranking from a classifier that depends entirely on the model class, in other words, it depends on the way the model is built.
If we choose to use the proportion of positive point for each segment (leaf) as the score, and we then put all the points in one region on the same level in the ranking. This means that some pairs of points will have the same score, and this can lead to ties in the ranking. Then, for large datasets, these regions will not contribute much to the total curve.

6.4.9 Precision-Recall Curve

The Precision-Recall curve is another way to evaluate the performance of a classification model, similar to the ROC curve, but in the axis we have precision and recall instead of TPR and FPR. The area under the Precision-Recall curve (AUC-PR) can be used as a metric to compare different models, especially in cases of class imbalance, where the ROC curve can be misleading.
However in some cases, the Precision-Recall curve can be more difficult to interpret than the ROC curve, since it is not normalized and it can be affected by the class imbalance in a different way than the ROC curve. Both curves can be useful in different scenarios, it's suggested to plot both.

6.4.10 Use of Metrics

There are many others metrics that can be used to evaluate classification models, and they can be applied to all the sets (training, validation, and test set). For instance, if we compute the accuracy on the test set, we are talking about the **test accuracy**, used to show if our conclusions are valid. Meanwhile, the **validation accuracy** is used to tune the hyperparameters of the model. And

the **training accuracy**, which is used in comparison with the validation accuracy (difference **generalization gap**) to understand if the model is overfitting or underfitting the data. It's important to note that it's not used for evaluating on the training set.

6.5 No Free Lunch Theorem

A question that often aries is: which classifier, model, search methods is the best? The aim is to make a guess before seeing data, but the No Free Lunch Theorem states that is not possible.

If you average over every possible problem, no optimization or learning algorithm is better than any other. An algorithm only looks good when the problem happens to match its assumptions. This means that there is no universally best model or algorithm. (**problem of induction**)

The **inductive bias** of a learning algorithm or model are those assumptions about the domain that are explicitly or implicitly hardcoded into the model. In other words, the inductive bias is the set of assumptions that a model makes about the data in order to generalize from the training data to unseen data. As a computer scientist, we need to figure out which of the many possible inductive biases is the best for our problem. For instance a linear model has a strong inductive bias towards linear relationships between features and the target variable.

Finally, when two models explain the data equally well, we usually prefer the simpler one. The intuition is that simple patterns are more likely than very complex ones, so learning methods often include a built-in **bias toward simplicity**.

7 Experiment Design

7.1 Introduction

Any measurement, without knowledge of the uncertainty, is **meaningless**. Here, "meaningless" means a measurement lacks reliability if its uncertainty is unknown; without knowing its precision or accuracy, the value cannot be trusted or interpreted and may be **misleading**.

Experimental measurements are essential to understand system behavior, allowing us to:

- Decide to **ship or not**
- **Identify** improvements for next iterations
- **Communicate** system behavior to others

This enables estimation of system performance for properties and conditions of interest.

We focus on experimenting rather than just evaluating or testing because:

- We always use a data sample, lacking access to all or future data
- Results vary due to factors beyond the dataset (workflow, randomness, etc.)

We can experiment with various LLM system components:

- **Document Base**: quality of the knowledge base or corpus
- **Retrieval Algorithm**: effectiveness in finding information
- **LLM Answer Generation**: quality of answers from context

Isolating what to test is harder than in traditional software due to interdependent components and less control. Some measures are end-to-end (whole system), others pinpoint where to intervene.

If done and reported correctly, more experiments provide more information; otherwise, they introduce bias.

Before starting, define uncertainty using basic probability principles.

7.2 Basic, Probabilities and Random Variables

7.2.1 Definition of Probability

From a general perspective, probability is a number between 0 and 1 to which we assign a meaning. A frequentist defines it as "*A proportion (# of times an event occurs)/repetitions as experiments repetitions tends to infinity*". However, this approach cannot work with us, since we cannot repeat experiments infinitely, and we cannot experiment always in the same conditions.

So, more generally the definition on which we are interested is "*A belief/likelihood of an event occurring*" or "*a belief on the distribution of a random variable*".

More formally we define:

- **Sample Space (S)**: The set of all possible outcomes of an experiment.
- **Outcome**: An outcome is a single element of the sample space.
- **Event (E)**: An event is a subset of the sample space that satisfies some conditions.

We define $P(E)$ as the probability of an event E occurring, and we have the following axioms:

More formally, we can define the two interpretations of probability as follows:

- **Frequentist Interpretation**:

  $$P(E) = \lim_{n \to \infty} n(E)/n$$

  where $n(E)$ is the number of times event E occurs in n trials.

- **Bayesian Interpretation**:

  $$P(E) = P(E|I) = belief$$

  where I is the background information and belief is a subjective degree of confidence in the occurrence of event E. This can be seen as an estimation of the frequentist probability, where the belief is updated as new evidence is obtained, following Bayes' theorem.

7.2.2 Axioms of Probability

For both interpretations, we have the following axioms of probability:

- **Non-negativity**: $P(E) \geq 0 \forall E \in S$.

- **Normalization**: $P(S) = 1$.
- **Complement Rule**: $P(E') = 1 - P(E)$, where E' is the complement of E.

Then if an event E is **mutually exclusive** with another event F (i.e., they cannot occur simultaneously), we have:

$$P(E \cup F) = P(E) + P(F) \tag{1}$$

This is the simplest case, and can be extended to more than two events.

In the case we have all outcomes **equally likely**, we can calculate the probability of an event E as:

$$P(E) = \frac{|E|}{|S|} \tag{2}$$

And works only if we have a finite sample space, and all outcomes are equally likely.

7.2.3 Random Variables

We define a **random variable** as a function that maps each outcome of an experiment to a real number, capturing uncertainty. Assigning a value to a random variable defines an event, and we can describe its behavior with a probability distribution.

For example, in a coin toss, let X be 1 for heads and 0 for tails. This is a **binary random variable** since it takes two values. If both outcomes are equally likely, the probability distribution is:

$$P(X = 1) = 0.5, \quad P(X = 0) = 0.5 \tag{3}$$

Probability Mass Function (PMF) The **probability mass function (PMF)** of a discrete random variable gives the probability that it equals a specific value: $P(X = k)$ for each possible k. The PMF represents the distribution and allows calculation of event probabilities.

7.3 Foundational Random Variables

Some random variables are classic and foundational in probability theory. They can be used to model various phenomena and are essential for understanding uncertainty in experiments.

7.3.1 Binomial Random Variable

The scenario we are describing is the following:

- n independent trials
- A consistent probability of success p in each trial, binary outcomes (success/failure)
- We want the probability of observing k successes in those n trials

The probability of observing k successes in n trials is given by the **binomial distribution**:

$$X \sim Bin(n, p) \tag{4}$$

Where X is the random variable representing the number of successes, and its PMF is:

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k} \tag{5}$$

Where $\binom{n}{k}$ is the binomial coefficient, calculated as:

$$\binom{n}{k} = \frac{n!}{k!(n-k)!} \tag{6}$$

The binomial distribution is useful for modeling scenarios like coin tosses.

If we plot the PMF of a binomial distribution with $n = 10$ and $p = 0.5$, we would see a bell-shaped curve centered around $k = 5$, indicating that observing 5 successes is the most likely outcome.

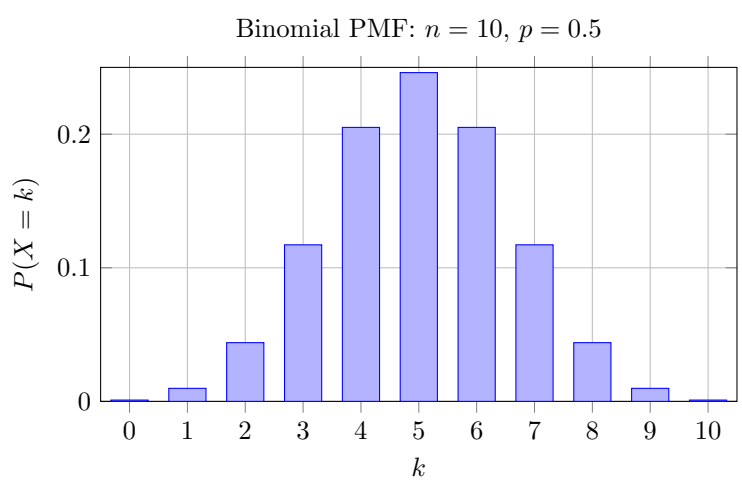

Figure 1: PMF of the binomial distribution with $n = 10$, $p = 0.5$

7.3.2 Bernoulli Random Variable

The Bernoulli distribution is a special case of the binomial distribution where $n = 1$. It models a single trial with two possible outcomes: success (1) with probability p and failure (0) with probability $1 - p$. The PMF of a Bernoulli random variable X is:

$$P(X = 1) = p, \quad P(X = 0) = 1 - p \tag{7}$$

7.3.3 Other PMFs Random Variables

Other important discrete random variables include:

- **Geometric**: Models the number of trials until the first success in a sequence of independent Bernoulli trials.

$$P(X = k) = (1 - p)^{k-1} p$$

- **Negative Binomial**: Models the number of trials until a specified number of successes occurs.

$$P(X = k) = \binom{k-1}{r-1} p^r (1 - p)^{k-r}$$